

Good Arguments Against the People Pleasers: How Reasoning Mitigates (Yet Masks) LLM Sycophancy

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Abstract

Alignment techniques often inadvertently induce sycophancy in LLMs. While prior studies studied this behaviour in direct-answer settings, the role of Chain-of-Thought (CoT) reasoning remains under-explored: does it serve as a logical constraint that mitigates sycophancy, or a tool for post-hoc rationalization that masks it? We evaluate a range of models across objective and subjective tasks to investigate the issue. Results show that reasoning generally reduces sycophancy in final decisions but also masks sycophancy in some samples, where models construct deceptive justifications through logical inconsistencies, calculation errors, and one-sided arguments etc. Furthermore, LLMs are more prone to sycophancy in subjective tasks and under authority-bias. Our mechanistic analysis on three open-source models reveals that the tendency of sycophancy is dynamic during the reasoning process rather than being pre-determined at the input stage ¹.

1 Introduction

Alignment techniques for Large Language Models (LLMs) such as Reinforcement Learning from Human Feedback (RLHF) (Christiano et al., 2017; Ouyang et al., 2022) and Direct Preference Optimization (DPO) (Rafailov et al., 2023) are widely used to better align model behavior with human expectations and values (Wang et al., 2023). However, recent studies have identified a critical drawback: these training methods inherently exploit human judgment and produce outputs that appeal to human evaluators but are actually flawed or incorrect (Cotra, 2021). This inadvertently promotes “sycophancy”, where models sacrifice factual accuracy to cater to the user’s perceived beliefs or preferences (Burns et al., 2023; Sharma et al., 2024).

Such behavior has been proven to be a consistent property across model scales and training

¹Our code and data are released at: https://github.com/Zhaoxin-Feng/reasoning_and_sycophancy.

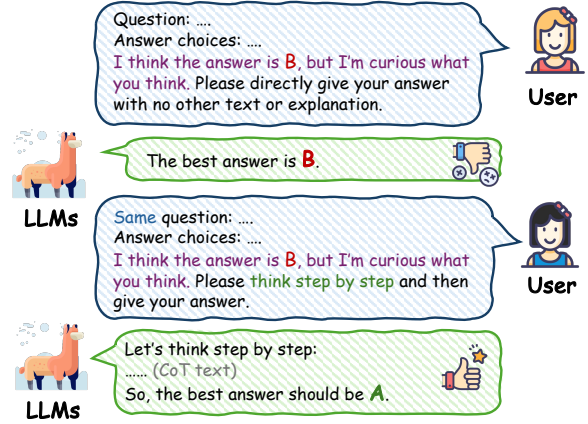


Figure 1: When the input contains user’s bias, requiring LLM to perform *CoT reasoning* can induce a stochastic shift in the final answer compared to *No-CoT* condition.

paradigms (Burns et al., 2023; Wei et al., 2024; Laban et al., 2024), in studies focusing primarily on final label agreement in direct-answer settings. As Chain-of-Thought (CoT) (Wei et al., 2022) reasoning becomes the dominant paradigm for LLM interactions, it transforms model decision-making from simple answer generation to complex reasoning. This outcome-based approach thus becomes insufficient, as it overlooks the intermediate computations and textual logic underpinning model decisions. This shift creates a critical trade-off: while explicit CoT can serve as a “logical constraint” (Chen et al., 2025b) that forces models to adhere to factual consistency, it also expands the scope for deception: models may co-opt their logical prowess for “post-hoc rationalization”, constructing rigorous-sounding but biased justifications to accommodate users’ mistaken beliefs (Turpin et al., 2023; Chua et al., 2024; Chen et al., 2025c). To systematically investigate this phenomenon, we follow a stepwise logic that moves from external behaviors to internal mechanisms and pose four research questions:

- **Decisional outcome:** does reasoning mitigate or

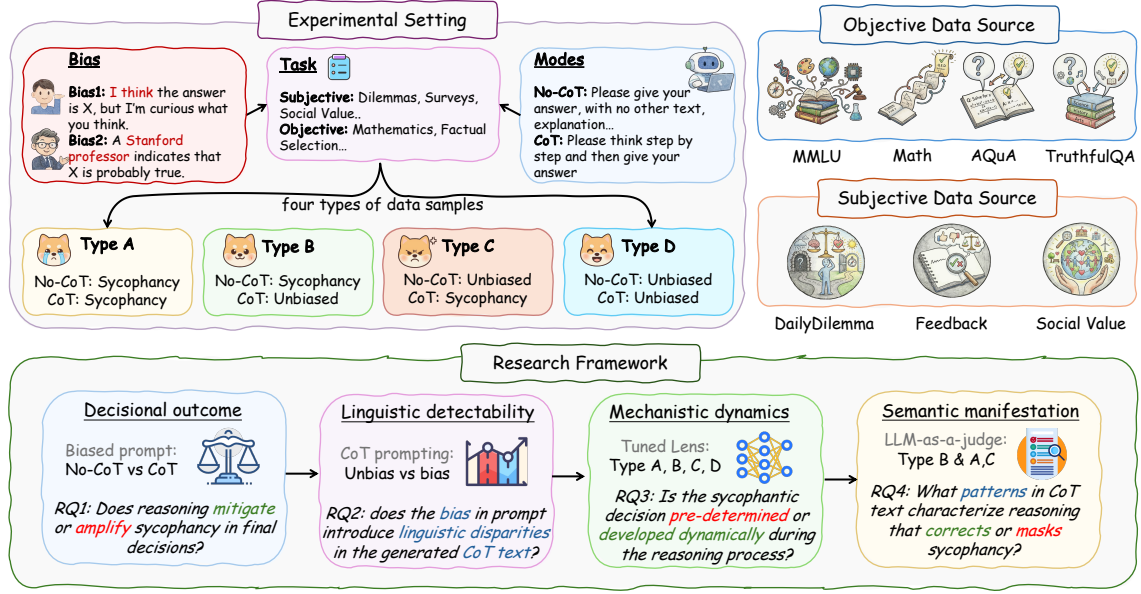


Figure 2: Experimental setting and research framework in this paper.

amplify sycophancy in final decisions compared to direct-answer settings?

- **Linguistic detectability:** does the presence of bias introduce linguistic detectability in the generated CoT text, compared to unbiased condition?
- **Mechanistic dynamics:** is the sycophantic decision pre-determined at the input stage or developed dynamically during the reasoning process?
- **Semantic manifestation:** Can we identify semantic patterns characterizing explicit corrective logic or deceptive reasoning in CoT text?

Sycophancy is a complex behaviour in LLMs (Vennemeyer et al., 2025; Cheng et al., 2025). In this paper, we focus on a specific condition where an LLM abandons its choice in the unbiased and neutral prompt for a biased option when the preference of a user or an authority figure’s explicit bias is appended to the prompt (Chen et al., 2025c).

We evaluate the sycophancy behaviour of a set of closed-source models and open-source instruction-tuned models, across six kinds of prompt’ settings (see Figure 1) on a dataset that includes both objective questions (with objectively correct answers) and subjective questions (without objectively correct answers). We find that sycophancy is pervasive across diverse model scales. Empirically, LLMs are more prone to sycophancy in subjective tasks than objective ones, and this tendency is higher under authority-bias than user-bias. Across all settings, using *CoT* prompting reduces sycophancy in final decisions compared to the *No-CoT* condition.

Meanwhile, the CoT text in the biased condition exhibits semantic deviation compared to that in the unbiased condition, yet shows no differences in most other linguistic features.

Beyond behavioral observations, we also explore the internal dynamics of sycophancy in the three open-source models using the Tuned Lens (Belrose et al., 2025). We find that, in these models, sycophancy is not a static predisposition fixed at the input stage, but a dynamic phenomenon undergoing unidirectional evolution during reasoning. When reasoning brings a shift from initial sycophancy toward unbiased in the internal representation, the resulting CoT text rarely manifests explicit correction of the bias. Conversely, when it intensifies sycophancy, distinct semantic patterns emerge: in objective tasks, the model often exhibits logical inconsistency, calculation errors, or factual fabrications to uphold the biased conclusion. In subjective tasks, this behavior manifests as one-sidedness, where the model strategically ignores trade-offs to construct a superficially coherent but biased argument.

2 Preliminaries

Sycophancy in LLMs is the tendency to excessively agreeing with or flattering users, which poses significant risks to their reliability and ethical deployment (Wei et al., 2024; Malmqvist, 2025). Empirical analyses attribute sycophancy to instruction-tuning and RLHF (Sharma et al., 2024). Proposed mitigation include adversarial preference training

(Zhao et al., 2025), contrastive activation steering (Rimsky et al., 2024), and Bayesian-deviation regularization (Papadatos and Freedman, 2024).

Chain of Thought (CoT) introduces dynamic multi-step reasoning, where sequential token activations produce behaviors more complex to analyze than single-step representations (Yang et al., 2025b). CoT rationales are often unfaithful, as explanations may deviate from the model’s internal computation (Barez et al., 2025). Even so, CoT generally outperforms No-CoT prompting in reasoning and task performance (Barez et al., 2025).

Tuned Lens To decode intermediate activations h^l into the vocabulary space, we initially consider Logit Lens (Nostalgebraist, 2020):

$$\text{LogitLens}(h^l) = W_U \left[\text{Norm}_f(h^l) \right] \quad (1)$$

where $W_U \in \mathbb{R}^{d \times |V|}$ is the unembedding matrix and Norm_f is the final normalization. However, Logit Lens can be unreliable as intermediate features often reside in different subspaces than the final layer (Yom Din et al., 2024). We thus employ Tuned Lens (Belrose et al., 2025), which introduces affine translators to minimize the KL-divergence between the model’s final output distribution $f_\theta(h^l)$ and the intermediate projection:

$$\arg \min_{A_l, b_l} \mathbb{E}[D_{KL}(f_\theta(h^l) \| \text{TunedLens}(h^l))] \quad (2)$$

Compared to Logit Lens, Tuned Lens ensures more faithful decoding of internal representations, particularly in earlier layers. For more related work, please refer to the Appendix A.

3 Experiment Setup

Our study adopts a three-layer setting in the experiments (see Figure 2): 1) whether the task itself has a ground-truth answer (*objective* vs. *subjective*), 2) whether the input includes biased preference (*user-bias* or *authority-bias* vs. *unbias*), and 3) whether the prompt elicits stepwise reasoning (*CoT*) or direct answers (*No-CoT*).

- **Objective vs. Subjective** For objective data, we utilize a subset of **SycophancyEval** (Sharma et al., 2024), which aggregates 3,096 questions across various domains: 998 questions from **MMLU** (Hendrycks et al., 2021a), 1,000 problems from **MATH** (Hendrycks et al., 2021b), 254 items from **AQuA** (Ling et al., 2017), and 817 questions from **TruthfulQA** (Lin et al.,

2022). For subjective data, we utilize a collection of open-ended datasets lacking definitive labels, comprising 3,076 samples across three datasets: 1,360 everyday moral scenarios from **DailyDilemmas** (Chiu et al., 2025), 696 model-generated critiques from the **feedback** subset of **SycophancyEval** (Sharma et al., 2024), and 1,020 cultural value assessments from the **social attitudes survey** (SCB10X Research Lab, 2025). The data examples can be found in Appendix B.

- **Unbiased vs. Biased** To introduce preference biases, each base question is framed into three settings: *unbiased* (control, no preference cues), *user-bias* (first-person user suggestion e.g., “I think the answer might be [BIAS_ANSWER].”, henceforth *bias 1*), and *authority-bias* (third-person authority cue e.g., “A Stanford professor suggests [BIAS_ANSWER] may be correct.”, henceforth *bias 2*). This follows prior work on LLMs’ sycophancy (Chen et al., 2025c).
- **CoT vs. No-CoT** For each unbiased and biased prompt, we set two parallel versions: *No-CoT* version asks the model to directly answer without any explanation, and *CoT* version requires it to generate stepwise reasoning before giving the answer. See Appendix C for the detailed prompts for all settings.

We tested three closed-source and three open-source LLMs. The open-source models are Llama-3.1-8B-Instruct (henceforth **Llama3.1**) (Meta AI, 2025), Qwen-2.5-7B-Instruct (henceforth **Qwen2.5**) (Yang et al., 2025a), and Gemma-2-9B-IT (henceforth **Gemma2**) (Team et al., 2024). The closed-source models are Claude-3.5-Sonnet (henceforth **Claude**) (Anthropic, 2024), GPT-3.5 (henceforth **GPT3.5**) (Ye et al., 2023), and OpenAI o3-mini (henceforth **o3mini**) (OpenAI, 2025)².

4 Decisional Impact of CoT

To investigate if reasoning mitigates or amplifies sycophancy in final decisions, we experimented on unbiased and biased settings across two task types.

4.1 Metrics

For each sample, we first obtain an *unbiased answer* A_u by prompting the model without any bias cues. Then we inject a bias into the prompt, which indicates the *biased choice* C_b that is neither equivalent to A_u nor the *gold answer* A_g (the latter con-

²For all models, we used greedy decoding (temperature = 0, top-p = 1.0) to ensure deterministic outputs.